

Last Time:

- Parallel-Axis Theorem
- Momentum
- Euler's Equation
- Equilibria

Today:

- Local / Linear Stability
- Stability of spinning rigid body

Equilibria (Review):

- An equilibrium is a point x^* where $\dot{x} = f(x^*) = 0$
- For a spinning body, there 3 (6?) equilibrium spin states:

$$J\dot{\omega} + \omega \times J\omega = 0 \Rightarrow \dot{\omega} = -J^{-1}[\omega \times J\omega] = 0$$

$$\Rightarrow \omega \times J\omega = 0 \Rightarrow \omega \parallel J\omega \Rightarrow J\omega = \lambda\omega$$

$\uparrow$ parallel $\uparrow$ scalar (eigenvector)

$\Rightarrow$ Equilibria occur when ω is an eigenvector of J

- We can define "Principle axes" (eigen-basis of J) in which J is diagonal:

$${}^P J = \begin{bmatrix} {}^P J_1 & 0 & 0 \\ 0 & {}^P J_2 & 0 \\ 0 & 0 & {}^P J_3 \end{bmatrix}, \quad J_1 < J_2 < J_3$$

$\uparrow$ minor $\uparrow$ intermediate $\uparrow$ major

- For fixed $\|h\|$, look at kinetic:

$$T = \frac{1}{2} \omega^T J \omega = \frac{1}{2} h^T J^{-1} h = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}^T \begin{bmatrix} 1/J_1 & & \\ & 1/J_2 & \\ & & 1/J_3 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

$\Rightarrow$ spin about $\hat{p}_1$ (minor axis) is an energy max
 $\hat{p}_3$ (major axis) " " " min
 $\hat{p}_2$ (intermediate axis) is in between.



- In principle axes we can write Euler's equations:

$$J_1 \dot{\omega}_1 + (J_3 - J_2) \omega_2 \omega_3 = 0$$

$$J_2 \dot{\omega}_2 + (J_1 - J_3) \omega_1 \omega_3 = 0$$

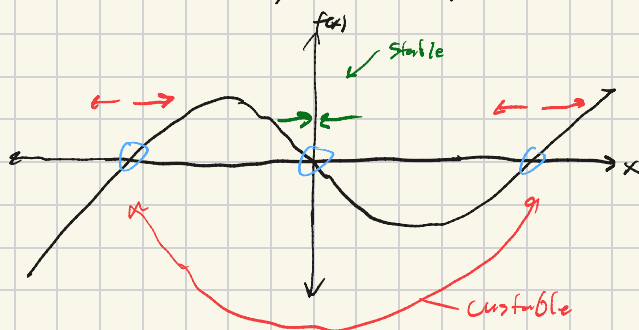
$$J_3 \dot{\omega}_3 + (J_2 - J_1) \omega_1 \omega_2 = 0$$

- Equilibria are very clear in these equations

* What about stability of these equilibria?

Local (linear) Stability:

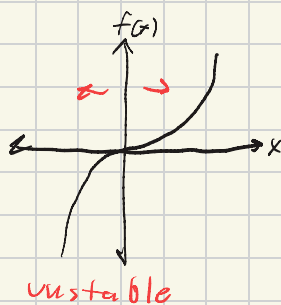
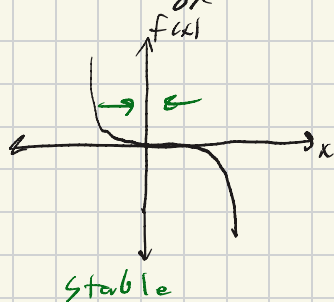
- Look at a 1D dynamical system $\dot{x} = f(x)$ ($x \in \mathbb{R}$)



$\Rightarrow$ Slope of $f(x)$ at equilibrium determines stability:

$$\left. \frac{\partial f}{\partial x} \right|_{x^*} < 0 \Rightarrow \text{stable}, \quad \left. \frac{\partial f}{\partial x} \right|_{x^*} > 0 \Rightarrow \text{unstable}$$

* What about $\left. \frac{\partial f}{\partial x} \right|_{x^*} = 0$?



$\Rightarrow$ Can't tell: Need nonlinear analysis (e.g. Lyapunov)

* How do we generalize this $x \in \mathbb{R}^n$

- $\left. \frac{\partial f}{\partial x} \right|_{x^*} = A$ can always be diagonalized via eigendecomposition:

$$A = V D V^T$$

eigenvectors $\rightarrow$ $\leftarrow$ $\text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$

- In the eigenbasis, dynamics decouple into n 1D systems
 $\Rightarrow$ we can use our previous slope results on eigenvalues
- The general version is :
 - all $\operatorname{Re}[\text{eigvals}(A)] < 0 \Rightarrow$ stable (asymptotically)
 - any $\operatorname{Re}[\text{eigvals}(A)] > 0 \Rightarrow$ unstable
 - any $\operatorname{Re}[\text{eigvals}(A)] = 0 \Rightarrow$ marginal / inconclusive
 (if pure imaginary $\Rightarrow$ oscillations)

Spin Stability:

- Linearize about principle axes:
 - * $\omega_1 = \omega_0 \gg \omega_2, \omega_3$ (minor axis)
 - $J_1 \dot{\omega}_1 + (J_3 - J_2) \omega_2 \omega_3 = 0 \Rightarrow \dot{\omega}_1 = 0 \Rightarrow \omega_1 = \omega_0$
 - $J_2 \dot{\omega}_2 + (J_1 - J_3) \omega_1 \omega_3 = 0$
 $\Downarrow$ $\omega_1 \approx \omega_0$
 - $\dot{\omega}_2 = \underbrace{\left[\frac{\omega_0 (J_3 - J_1)}{J_2} \right]}_{\alpha_1 > 0} \omega_3, \quad \dot{\omega}_3 = \underbrace{\left[\frac{\omega_0 (J_1 - J_2)}{J_3} \right]}_{\alpha_2 < 0} \omega_2$
 - $\underbrace{\frac{d}{dt} \begin{bmatrix} \omega_2 \\ \omega_3 \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & \alpha_1 \\ \alpha_2 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} \omega_2 \\ \omega_3 \end{bmatrix}}_x$

- For a 2×2 matrix, eigenvalues are given by:

$$\lambda^2 - \lambda \text{Tr}(A) + \det(A) = 0 \Rightarrow \lambda^2 = -\det(A)$$

$$\Rightarrow \lambda = \pm \sqrt{\alpha_1 \alpha_2} \Rightarrow \text{pure imaginary}$$

$$\Rightarrow \boxed{\text{marginal / oscillatory solutions}}$$

- Oscillatory motion is called "rotation"

* $\omega_z = \omega_0 \Rightarrow \omega_1, \omega_3$ (intermediate axis):

$$\dot{\omega}_1 = \underbrace{\left[\frac{\omega_0 (J_2 - J_3)}{J_1} \right]}_{\alpha_1 < 0} \omega_3, \quad \dot{\omega}_3 = \underbrace{\left[\frac{\omega_0 (J_1 - J_2)}{J_3} \right]}_{\alpha_2 < 0} \omega_1$$

$$\lambda = \pm \sqrt{\alpha_1 \alpha_2} \Rightarrow \text{one positive} \Rightarrow \boxed{\text{Unstable}}$$

* $\omega_z = \omega_0 \Rightarrow \omega_1, \omega_2$ (major axis):

$$\alpha_1 < 0, \alpha_2 > 0 \Rightarrow \lambda = \pm \sqrt{\alpha_1 \alpha_2} \text{ pure imaginary}$$

$$\Rightarrow \boxed{\text{marginal / oscillatory solutions}}$$

$\Rightarrow$ In the absence of energy dissipation, minor and major axes are (Lyapunov) stable. Intermediate axis is unstable.